Smart Meeting Memory Agent

COMPLETE TECHNICAL DOCUMENTATION & STEP-BY-STEP USER GUIDE

1. Executive Summary & Purpose

Professionals attend multiple meetings daily with clients, managers, and cross-functional teams. Critical decisions, deadlines, assigned commitments, and unresolved concerns are often forgotten or scattered across raw notes. Traditional AI chatbots operate on single sessions and do not retain context across interactions, forcing users to repeat background context before every meeting.

Smart Meeting Memory Agent solves this problem by acting as a persistent AI long-term memory system powered by **Hindsight** and **Google Gemini API**. It automatically extracts key facts, updates commitment statuses over time, isolates client memory contexts, and synthesizes structured executive briefings before upcoming meetings.

2. System Architecture & Tech Stack

Component	Technology Stack	Key Function & Purpose
Frontend UI	React 18, TypeScript, Vite, Tailwind CSS v4, Lucide Icons	Responsive SaaS dashboard, interactive demo controls, modals, and real-time memory feeds.
Backend API	Node.js, Express.js (Port 5050), TypeScript	REST API routing, extraction pipeline, briefing generator, SQLite database controllers.
Database Layer	SQLite via sql.js (WASM + Atomic Disk Persistence)	Zero-config relational database storing meetings, extracted items, commitments, and settings.
Long-Term Memory	**Hindsight Memory System** (hindsightService.ts)	Persistent semantic fact retention, temporal status updates, and strict client isolation.
AI Provider	Google Gemini 2.5 Flash API + Heuristic Engine	Parses meeting notes, extracts structured facts, generates briefings, and answers Q&A; chat.

3. Hindsight Memory Architecture Deep Dive

Hindsight serves as the primary long-term memory engine. Unlike standard chatbots that append messages into a temporary window, Hindsight structures memories into discrete categories:

- **Client Preferences**: Technical requirements, preferred modes, delivery styles (e.g. *'Client requested Dark Mode'*).
- **Decisions**: Formal agreements reached during meetings (e.g. *'Approved payment architecture B'*).
- **Commitments**: Tasks promised by team members or clients with assigned owners and deadlines.
- **Action Items**: Operational follow-ups and task assignments.
- **Unresolved Concerns**: Open issues, security risks, or pending questions raised by the client.

- **Historical Context**: Preserved records of past discussions.

Dynamic Memory Lifecycles: When a new meeting is recorded, Hindsight inspects existing active memories. If new notes indicate a previously unresolved concern or task has been addressed, Hindsight updates the status of the older memory item to resolved or completed while preserving the historical record. Contexts for different clients are strictly isolated.

4. Exhaustive Feature Guide & Step-by-Step Task Instructions

Feature 1: Dashboard Overview

The Dashboard is the executive command center displaying top KPI metrics (Total Meetings, Active Commitments, Completed Tasks, Overdue Tasks, Open Concerns), recent meeting highlights, upcoming deadlines, and a live feed of memories recorded in Hindsight.

Step-by-Step Task Instructions:

1. Click **Dashboard** in the sidebar navigation.
2. Click any KPI card (e.g. *Active Commitments*) to jump directly to that filtered view.
3. Click **'Prepare Me Now'** on the top purple banner to generate an immediate briefing.
4. Click **'+ Add Meeting'** to open the meeting note recording drawer.

Feature 2: Meeting Capture & AI Information Extraction

Allows users to input meeting notes or transcripts. When submitted, the backend AI model automatically parses text and extracts Confirmed Decisions, Commitments (task, assigned owner, deadline), Action Items, Unresolved Concerns, Client Preferences, and Follow-up Requirements.

Step-by-Step Task Instructions:

1. Click **'+ Add Meeting'** in the top navigation bar or **'Record New Meeting'** on the Meetings view.
2. Enter **Meeting Title** (e.g. *'ABC Client Payment Sync'*) and **Client Name** (e.g. *'ABC Client'*).
3. Paste raw meeting notes or click **'Load Sample Meeting Notes'** to auto-populate sample text.
4. Click **'Process & Save Meeting'**.
5. Click **'View Details & Extraction'** on the saved meeting card to inspect AI extracted items.

Feature 3: Searchable Meeting History

Provides a centralized repository of all historical meetings with client pill filtering and full-text keyword searching.

Step-by-Step Task Instructions:

1. Click **Meetings** in the sidebar.
2. **Filter by Client**: Click any client pill (e.g. *'ABC Client'*) at the top.
3. **Search Notes**: Type key terms (e.g. *'security'*, *'dark mode'*) into the search bar.
4. **View Details**: Click **'View Details & Extraction'** to view notes alongside extracted facts.
5. **Delete Meeting**: Click the trash icon on a card to delete it.

Feature 4: Prepare Me for My Next Meeting (Briefings)

Flagship feature. Select a client, and the AI queries Hindsight long-term memories, commitments, and past meetings to generate a structured executive briefing containing Previous Meeting Summary, Confirmed Decisions, Tracked Commitments & Status, Unresolved Concerns, Suggested Questions, Important Reminders, and a Natural-Language Memory Synthesis beginning with *'Here is what I remember about [Client]...'*.

Step-by-Step Task Instructions:

1. Click **Prepare for Meeting** in the sidebar.

2. Select the target client from the dropdown menu (e.g. *'ABC Client'*).
3. Click **'Generate Briefing'**.
4. Read the top purple **Hindsight Synthesis Card** for natural-language memory recollection.
5. Review **Tracked Commitments** and **Suggested Questions** to frame your meeting agenda.

Feature 5: Commitment & Task Tracking

Dedicated task dashboard tracking task description, client, assigned owner, deadline, and status (Pending, In Progress, Completed, Overdue). Updating a task status automatically syncs back to Hindsight memory!

Step-by-Step Task Instructions:

1. Click **Commitments** in the sidebar.
2. **Filter by Status**: Click tabs (*All*, *Pending*, *In Progress*, *Completed*, *Overdue*).
3. **Update Status**: Click the status dropdown badge on any task card (e.g. change *Pending* to *Completed* ✓).
4. Notice that marking a task completed updates its status in Hindsight memory for future briefings.

Feature 6: AI Assistant Conversational Chat

Interactive AI chatbot where users can ask questions about past meetings, client requests, promises, or deadlines using retrieved Hindsight memories.

Step-by-Step Task Instructions:

1. Click **AI Assistant** in the sidebar.
2. (Optional) Filter chat context by selecting a specific client from the top right dropdown.
3. Click any **Sample Question** button (e.g. *'What did Ravi promise?'*) or type a custom question.
4. Press **Send** to view factual answers with memory citations.

Feature 7: Hindsight Memory Explorer

Visual inspection tool allowing users to view, edit, or delete memories stored in Hindsight across categories (Preferences, Decisions, Commitments, Action Items, Unresolved Concerns, Historical Context). Edits directly shape future AI responses!

Step-by-Step Task Instructions:

1. Click **Memory Explorer** in the sidebar.
2. Click category pills to filter memories.
3. **Edit Memory**: Click the pencil icon on a card, edit text/category, and click checkmark.
4. **Delete Memory**: Click the trash icon to remove outdated or incorrect facts.

Feature 8: Hackathon 8-Step Interactive Demo Walkthrough

A top banner allowing evaluators to test all core capabilities in an automated 8-step sequence.

Step-by-Step Task Instructions:

1. Click **'Step 1: Seed ABC Client Discovery Meeting'** on the top banner.
2. Click **'Step 2 & 3'** to inspect Hindsight memory explorer.

3. Click **'Step 4 & 5'** to generate the Prepare Me briefing for ABC Client.
4. Click **'Step 6'** to add a follow-up meeting resolving API docs.
5. Click **'Step 7'** to regenerate briefing and verify status updates.
6. Click **'Step 8'** to run the Client Isolation Test for XYZ Corp.

Feature 9: System Settings & Health Monitor

Displays health status of Express backend, Hindsight memory engine, and Gemini API provider with 1-click database seed and reset controls.

Step-by-Step Task Instructions:

1. Click **Settings & Demo** in the sidebar.
2. Review **Backend & Service Health** cards.
3. Click **'Preload Demo Data'** or **'Reset Database'** as needed.

5. Installation & Local Execution Guide

Follow these instructions to run the application on any local machine:

Step 1: Install Dependencies & Build Project

```
npm run setup
```

Step 2: Start the Application

```
npm start
```

Open your browser to: **http://localhost:5050** (Production) or **http://localhost:3000** (Dev).

6. Complete API Endpoint Reference

Method	Endpoint	Description & Purpose
POST	/api/meetings	Record meeting & process AI memory extraction.
GET	/api/meetings	List meetings (supports search & client filters).
GET	/api/meetings/:id	Fetch single meeting details & extracted data.
DELETE	/api/meetings/:id	Delete meeting record.
POST	/api/briefings/generate	Generate 'Prepare Me' client briefing.
POST	/api/chat	AI Chat query using retrieved Hindsight memory.
GET	/api/commitments	List commitments (filtered by client/status).
PUT	/api/commitments/:id	Update commitment status & sync Hindsight memory.
GET	/api/memories	Inspect all Hindsight memory items.
PUT	/api/memories/:id	Edit memory content, category, or status.
DELETE	/api/memories/:id	Delete memory item.
POST	/api/demo/seed	Preload hackathon sample dataset.
POST	/api/demo/reset	Reset database cleanly.
GET	/api/health	System health and Hindsight connection check.